GSOC 2026 PROPOSAL - FOSSASIA

Improve the Magic ePaper Badge app into a User friendly Functional App

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Personal Details and Contact Information

Hi! I am Jayasankarj, a first year student from the SRM AP University ,India. I am an avid open-source enthusiast who loves making (and breaking) software.

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TECHNICAL SKILLS

- Object-oriented programming skills

Data Structures:

Linear:

- Arrays
- Linked list
- Stack
- Queue

Non-Linear:

- Trees

Algorathims:

- Hashing
- Binary search
- Greedy
- Dynamic programming
- Backtracking

Web Development

- for UI and Ux
- frontend

- BackendDesign
- Interest in contributing to open source

Languages:

- C
- C++
- Java
- HTML
- CSS
- JavaScript

Commitment:

I really want to help the Magic ePaper Badge ecosystem in the long term and make it easier to use and better for contributors

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ABSTRACT

The Magic ePaper Badge is an innovative hardware platform. It features a tri-color ePaper display, NFC communication, low power usage, and portable badge

usability. This creates a unique interactive system for makers, developers, event communities, and hardware enthusiasts. Though the existing open-source Flutter companion app lays the groundwork for badge interaction, the user experience, workflow, editing tools, and NFC transfer reliability need significant improvements. This project aims to enhance the functionality and usability of the Magic ePaper Badge application, transforming it into a production-ready, contributor-friendly, and intuitive cross-platform solution. The main focus of this project is to redesign the application to support a modern user-friendly editing workflow. Users should easily create customized badge content using freehand drawing tools, text insertion, stickers, imported images, and visual effects within a smooth and responsive Flutter interface. A significant technical challenge is ensuring that all created content is optimized for the tri-color ePaper display, as standard RGB rendering techniques are not suitable. To solve this, the project will introduce an intelligent image processing and palette conversion pipeline that can manage contrast balancing, brightness adjustments, threshold tuning, image rotation, cropping, scaling, and color reduction tailored for the badge hardware. Another important goal is to improve the NFC-based content transfer pipeline. This pipeline connects the mobile or desktop application to the physical badge. The existing workflow will be refined to include stable device detection, reliable packetized data transfer, safe writing mechanisms, validation after transfer, and visible progress updates. These enhancements will make the transfer process reliable for real-world use. From a software engineering viewpoint, the proposal also highlights

the need for maintaining the codebase, modular design, and scalability for contributors. The Flutter application will be organized into clean layers for presentation, state management, rendering, and device communication, simplifying feature extensions for future contributors. Additionally, the project will implement thorough widget tests, integration tests for NFC workflows, automated release processes through CI/CD, improvements in platform packaging, contributor documentation, and architecture diagrams. By the end of the GSoC coding period, the final outcome will be a fully optimized, feature-rich, stable, and user-friendly Magic ePaper Badge application. It will significantly enhance both user creativity workflows and the long-term maintainability of the open-source project.

PROBLEM STATEMENT

The Magic ePaper Badge can be interacted with using the current application's basic features, but there are a few key areas that require improvement:

- UI requires more user-friendly workflows and improved usability.
- Richer editing support is required for text and drawing tools.
- NFC gearbox ought to be more dependable.
- Optimisation is required for tri-color ePaper image rendering.
- Cross-platform support requires stabilization

- Contributor onboarding and long-term maintainability are impacted by current bugs.
- These problems will be resolved by this project, which will also prepare the app for production.

PROPOSED FEATURES

1. Modern UI/UX Redesign

- Responsive layouts and a cleaner Flutter user interface
- Improved drawing, text, emoji, and image editing navigation
- Fast switching between colour palettes
- Redo/undo history stack
- Enhanced workflow for project save/load

2) Sophisticated Tools for Drawing and Editing

- Support for freehand drawings
- Control over brush size and opacity
- Shape tools (line, circle, and rectangle)
- Text insertion with alignment, style, and font size
- Emoji and sticker placement
- Layer-like object editing flow

3) Intelligent Image Enhancement for ePaper

- Bring in PNG, JPG, and WebP pictures
- Automatic tri-color palette conversion
- Adjusting contrast, brightness, and threshold
- Aspect lock, rotation, crop, and resize

- Prior to badge transfer, preview

4) Dependable Transfer of NFC Content

- Quicker device identification
- Data packet transfer that is stable
- Retry method for unsuccessful writes
- Progress indicator while synchronising
- After writing, transfer validation

5) Deployment Across Multiple Platforms

- Optimisation for iOS and Android
- Linux, macOS, and Windows builds
- When feasible, web preview compatibility
- CI/CD for releases using GitHub Actions

6) Documentation + Testing

- UI tool widget tests
- NFC workflow integration tests
- Setup instructions for contributors
- Module documentation and architecture diagrams
- Public relations guidelines and templates

Module-Level Design

- presentation/ → Flutter screens, widgets, routing, responsive layouts
- core/editor/ → canvas engine, brush logic, shapes, text,

- emoji placement core/rendering/ → tri-color conversion, palette logic,
- image scaling services/nfc/ → transfer manager, retry-safe writes,
- validation state/ → Riverpod/Bloc providers for editor state and transfer progress platform/ → Android/iOS/Desktop platform-specific integrations tests/ → widget tests,
- rendering golden tests, and NFC integration mocks

Technical Implementation

This part talks about how the suggested changes can be made to the actual repository. The code examples below are meant to be examples of how the project will be built, showing the engineering direction of the project while still allowing for mentor feedback during community bonding.

1. Editor State Controller

A centralized state management system will be designed to control the entire editor workflow.

Implementation Overview

- Keep a single state for:
- Strokes for drawing
- Text parts
- Emojis and stickers
- Images that were brought in
- Make sure that the UI and canvas rendering are in sync.
- For scalability, use structured state models.

Key Benefits

- Separation of UI and business logic
- Easier debugging and testing
- Smooth support for undo/redo operations
- Scalable for future features like layers

2. Undo and Redo Workflow

A command-driven history mechanism will be implemented to track all user actions.

Implementation Overview

- Every action a user takes (like drawing, texting, deleting, or moving) is saved as a command.
- Commands that work:
- Execution (take action)
- Reversal (undo action)

Key Benefits

- A reliable editing process
- Can grow to handle complex features like layers and grouping
- Makes the user experience much better
- Follows design patterns for editors that are common in the industry

3. Tri-Color Rendering Engine

A specialized rendering pipeline will convert editor content into a format compatible with tri-color ePaper displays.

Implementation Overview

Convert RGB images into:

- Black
- White
- Red (tri-color palette)

Apply:

- Brightness normalization
- Contrast adjustments
- Threshold mapping

- Generate a display buffer optimized for hardware

Key Benefits

- Ensures readability on ePaper screens
- Reduces rendering errors
- Produces accurate preview before transfer
- Minimizes data size for NFC transfer

4. Reliable NFC Transfer Layer

A robust service layer will handle communication between the app and the physical badge.

Implementation Overview

- Identifying devices and starting sessions
- Data transfer via packets
- Retry mechanism for failed transmissions
- Post-transfer validation
- Progress tracking for user feedback

Key Benefits

- Reliable data transfer
- Improved hardware compatibility
- Fault-tolerant communication
- Better user confidence during syncing

5. Image Optimization Pipeline

An image preprocessing pipeline will enhance imported images before rendering.

Implementation Overview

- Image normalization steps:
- Resize and crop
- Rotation and scaling
- Brightness and contrast tuning
- Convert images into tri-color compatible format
- Apply dithering or threshold techniques

Key Benefits

- Improved visual quality on ePaper
- Better readability of text and graphics
- Optimized output for real-world usage
- Consistent rendering across devices

IMPLEMENTATION PLAN

Getting to Know Each Other (Before Coding)

- Look into the architecture of a study repository.

- Set up the local environment
- Talk to your mentors about the roadmap.
- Fix one or two problems with the starter
- Set the final limits for the module

Phase 1 (Weeks 1–3)

- The basis for the UI redesign
- Change the screens and navigation
- Add a drawing toolbar
- Make state management better.

Phase2 (Weeks 4–6)

- Text editor with emoji support
- Tools for importing and changing images
- Engine for converting three colours
- Rendering a preview in real time
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Phase 3 (Weeks 7–9)

- Refactor NFC transfer
- Retry and validation flow
- More devices that work with it
- Better ways to report errors

Phase 4 (Weeks 10–12)

- Packaging that works on all platforms
- Automated CI builds
- Testing widgets and integrations
- Guides for contributors and documentation

Things to deliver

- Flutter app ready for production
- Drawing and text tools that are more advanced
- Transfer of NFC badges that works
- Engine for rendering three colours at once
- Pipeline for releasing on multiple platforms
- Complete documentation for contributors

Expected Effect

This project will make the Magic ePaper Badge platform much easier to use by making it easy for end users to create content that is visually appealing and trustworthy. It will also

make it easier for people who want to add to the repository in the future.
The final result will make it easier for both hardware fans and app developers to use the badge.